Oxford College of Emory University

Chapter 15: Monetary Policy

Part 2: Policy in Practice

ECON_OX 112 · Principles of Macroeconomics

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Roadmap

- Understand how he Fed chooses a policy.
- Explain the lower bound and QE.
- Understand long-run effects.

Inflation Target

Inflation target

A goal for inflation over time. Usually 2%, announced every year.

Why working with the inflation Target?

1. Transparency
2. Reduces uncertainty.
   - The federal reserve will implement their policies following this target.
   - Usually, the announcement will also include a range of operation. For instance: a 2% target, with a range of operation between 1.6% and 2.4%.

Federal Reserve Board: inflation objective · Federal Reserve Board: policy rules

The Taylor Rule

A formula to guide Monetary Policy decisions

r*: neutral real rate	π: inflation	π*: target	g: output gap
2%	4%	2%	−2%

- **The Formula: $i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5g$**
- **"i" is the target interest rate.**
- **Output below potential:** • Lowers the suggested rate.
- **Inflation above target:** • Raises the suggested rate.

Calculate i. Compare your reasoning with a partner.

$i = 2 + 4 + 1 - 1 = 6\%$

Federal Reserve Board: policy rules

The Zero Lower Bound

Usually, central banks cannot reduce interest rates far below 0.

Why is it a problem?

People can hold cash instead, and yet have a higher return. This can be understood as a limit for regular monetary policy.

The aftermath of crisis

After 2008 and during the pandemic, the central banks found their interest rates close to 0%. As a result, other policies had to be implemented.

Federal Reserve Board: policy limits

Quantitative Easing

What is this:

The central bank buys large amounts of longer-term bonds.

How it works?

Higher bond prices mean lower interest rates over the economy.

(similar to regular Monetary Policy) Lower borrowing costs can support C and I.

Federal Reserve Board: asset purchases and longer-term rates

Long Run Effects - Monetary Neutrality

Long run

Money supply increase changes prices, not real output, this phenomena is called Monetary Neutrality.

Real capacity

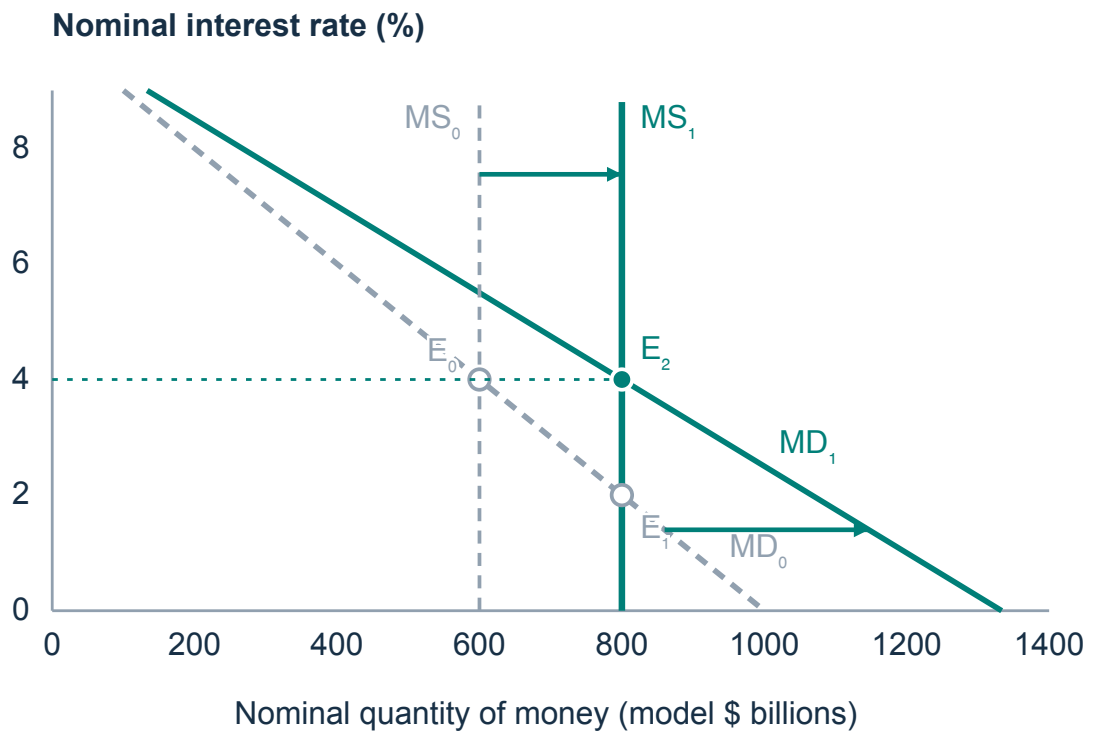
Recall, Workers, capital and technology determine potential output.

Money supply	Price level	Real output
+10%	+10%	Unchanged

Full price adjustment; other long-run conditions fixed.

ECB: short-run effects and long-run neutrality

Monetary Expansion



Start

MS = 600; i = 4%.

Short run

MS rises to 800; i falls to 2%.

Long run

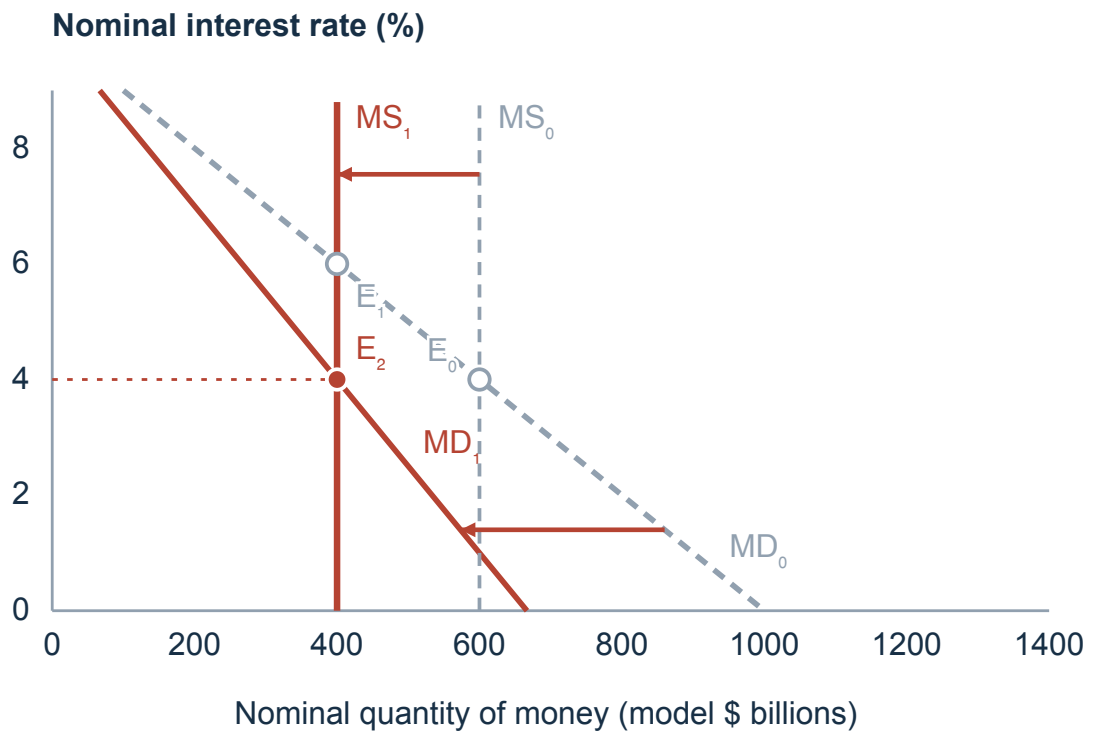
Prices rise. MD shifts right; i returns to 4%.

One-time MS change; long-run expected inflation unchanged.

New MS stays at 800; potential output is unchanged.

ECB: short-run effects and long-run neutrality

Monetary Contraction



Start

MS = 600; i = 4%.

Short run

MS falls to 400; i rises to 6%.

Long run

Prices fall. MD shifts left; i returns to 4%.

One-time MS change; long-run expected inflation unchanged.

New MS stays at 400; potential output is unchanged.

ECB: short-run effects and long-run neutrality

Long-Run Checkpoint

MS falls 20%. Prices fully adjust; other long-run conditions stay fixed.

Predict output, prices, and MD. Which curve moves first?

Real output	Price level	Nominal MD
Unchanged	−20%	Shifts left

MS moves first. Lower prices later shift MD left.

ECB: short-run effects and long-run neutrality